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Structure Learning–Based Interaction Feature Construction for Bank Failure Prediction

CHIWOO LEE[1], SUJIN PYO[2], and MINSU CHO[1,3], (Member, IEEE)
[1]School of Information Convergence, Kwangwoon University, Seoul 01897, South Korea
[2]Department of Industrial Engineering, Seoul National University, Seoul 08826, South Korea
[3]Department of Artificial Intelligence Applications, Kwangwoon University, Seoul 01897, South Korea

Corresponding author: Minsu Cho (E-mail: mcho@kw.ac.kr).

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ABSTRACT Early warning systems for bank failure prediction commonly rely on machine learning models trained on bank-level financial ratios. Although such models often achieve strong predictive performance, they offer limited insight into the joint signaling of distress by multiple indicators. This research introduces a dependency-guided interaction feature framework for bank failure prediction. The framework employs structure learning to identify dependency relationships among financial ratios and to transform them into edge-level interaction features. These features are then combined with the original ratios and incorporated into standard predictive models, without presuming causal validity. Empirical evaluation on a real-world bank failure dataset with severe class imbalance demonstrates that dependency-guided interaction features consistently enhance predictive discrimination and robustness when combined with original financial indicators. Furthermore, the explainability analysis indicates that the proposed representation facilitates relationship-level explanations, complementing traditional feature-level interpretations. This study contributes to establishing structure learning as a principled and model-agnostic feature-construction mechanism that enhances both predictive performance and interpretability in bank failure prediction.

INDEX TERMS Bank failure prediction, Structure learning, Financial risk modeling, Explainable machine learning, Early warning systems

I. INTRODUCTION

Early warning systems for bank failure prediction are a core component of financial supervision and risk management [1]. In practice, machine learning models for bank failure prediction are typically trained on bank-level financial ratios, including measures of profitability, leverage, liquidity, and capitalization [2]. While such models often achieve strong predictive performance, their explainability remains limited, particularly in terms of how multiple financial indicators jointly signal distress.

Most existing approaches treat financial ratios as independent input features [3]. As a result, post hoc explanations tend to highlight individual indicators, such as declining return on assets or rising leverage, without clarifying how interactions among ratios contribute to bank failure risk [4]. This limitation is problematic because bank distress rarely arises from a single ratio in isolation. Instead, failure is often associated with combinations of adverse conditions, such as weak profitability under high leverage or liquidity pressure interacting with inefficient asset utilization.

Although modern machine learning models can implicitly capture nonlinear interactions, they are not explicitly represented and are therefore difficult to interpret. Exhaustive interaction expansion, on the other hand, quickly becomes impractical and often degrades model stability. Consequently, there remains a gap between predictive accuracy and relationship-level explainability in bank failure prediction models [5].

Structure learning methods offer a promising yet underexplored opportunity in this context [6], [7]. By learning dependency relationships among financial ratios, these methods provide a structured representation of inter-feature associations. Without assuming causal validity, a learned dependency graph can be interpreted as a compact summary of statistically supported relationships, offering a principled basis for interaction modeling and explanation.

In this paper, we propose a graph-guided interaction feature framework for bank failure prediction. The framework first learns a dependency graph from bank-level financial ratios and then transforms each graph edge into an interaction feature that captures the instance-specific activation of a learned relationship. These interaction features are combined with original ratios and used in standard machine learning models. By constraining interaction modeling to learned dependencies, the proposed approach aims to improve both predictive performance and interpretability in terms of financial ratio relationships.

We evaluate the proposed framework on a real-world bank failure dataset characterized by severe class imbalance. Multiple structure learning algorithms and predictive models are compared under a unified experimental protocol, and randomization analyses are conducted to distinguish structured interaction effects from unstructured feature expansion.

The contributions of this study are threefold. First, we propose a dependency-guided interaction feature framework that bridges predictive performance and relationship-level explainability in bank failure prediction. By reframing structure learning as a feature-construction mechanism rather than a causal-discovery tool, the approach incorporates interdependencies among financial ratios without exhaustive interaction expansion. Second, through a unified experimental evaluation across multiple structure learning methods and predictive models, we show that dependency-guided interaction features consistently improve robustness and discrimination when combined with original financial ratios, while remaining largely insensitive to the choice of structure learning algorithm. Third, the framework enables relationship-level interpretation of bank failure risk, supporting transparent early warning signals and practical supervisory decision-making.

The remainder of this paper is organized as follows. Section II reviews related work on bank failure prediction, structure learning, and explainable machine learning. Section III introduces the proposed dependency-guided feature construction framework in detail. Section IV describes the experimental setup, including data, feature configurations, and evaluation metrics. Section V presents empirical results on predictive performance, robustness analyses, and explainability. Finally, Section VI discusses the implications, contributions, and limitations of the proposed approach, and Section VII concludes the paper.

II. RELATED WORKS

Machine learning–based approaches to bank failure prediction have been extensively studied as part of early warning systems for financial supervision. Early work relied on statistical models such as linear discriminant analysis and logistic regression, where financial ratios were manually selected and interpreted individually [8], [9]. These models typically used profitability, leverage, liquidity, and capitalization ratios as independent predictors of failure.

With advances in machine learning, more flexible nonlinear models have been introduced. For example, [10] employed logistic regression and tree-based models to predict distress in European banks, while [11] used random forests and boosted trees to identify large U.S. bank failures. Subsequent studies adopted gradient boosting machines and neural networks to capture nonlinear effects and improve predictive accuracy, particularly under severe class imbalance [12].

Despite their predictive success, these models generally treat financial ratios as independent input variables. Nonlinear interactions are learned implicitly within model architectures such as boosted trees or multilayer perceptrons, but they are neither explicitly represented nor directly interpretable. As a result, explanations typically focus on marginal feature importance, which obscures how combinations of indicators jointly contribute to bank failure risk. This limitation is fundamental, as bank distress rarely results from a single ratio in isolation. Instead, it typically emerges from conditional configurations, such as high leverage combined with weak profitability or inefficient asset utilization [13].

To improve transparency, recent studies have incorporated explainable machine learning techniques into bank failure and financial risk models. For example, [14], [15] applied SHAP-based explanations to tree ensembles and gradient boosting models, identifying the most influential financial ratios driving predicted risk. While these approaches enhance feature-level interpretability, they remain post hoc and do not explicitly model the dependency structure among ratios. Consequently, they offer limited support for relationship-level explanations, such as the idea that leverage risk depends on concurrent profitability or liquidity conditions.

Structure learning methods provide a complementary perspective by explicitly modeling dependencies among variables. Constraint-based algorithms such as the PC algorithm [6], score-based methods such as Greedy Equivalence Search [16], and continuous optimization approaches such as NOTEARS [17] and GOLEM [18] have been widely studied in the causal discovery literature. In financial research, these methods have mainly been used for causal inference, systemic risk analysis, or macro-financial network construction, often focusing on graph recovery or connectedness measures rather than predictive modeling [7].

The use of structure learning as a feature-construction mechanism for bank failure prediction remains largely unexplored. Existing studies rarely integrate learned dependency structures into downstream classifiers, nor do they systematically compare different structure learning paradigms within a unified predictive framework. Moreover, most applications emphasize causal interpretation, which is difficult to justify in observational supervisory data and may limit practical adoption.

III. METHOD

A. PROBLEM DEFINITION

Bank failure prediction is formulated as a binary classification problem at the bank level, with the objective of estimating the probability that a bank will fail within a predefined future horizon. Let the dataset be defined as

$$\mathcal{D} = \{(x^{(k)}, y^{(k)})\}_{k=1}^{n}, \tag{1}$$

where each observation corresponds to a single bank.

For bank k, the feature vector is given by

$$x^{(k)} \in \mathbb{R}^p, \tag{2}$$

where p denotes the number of standardized financial ratios. These ratios capture multiple dimensions of bank conditions, including profitability, leverage, liquidity, capitalization, and operational efficiency. These ratios are commonly used in supervisory stress testing and early warning systems, and they provide a snapshot of a bank's financial condition at a given point in time.

The target variable is defined as

$$y^{(k)} \in \{0, 1\}, \tag{3}$$

where $y^{(k)} = 1$ indicates that bank k experiences failure within the prediction horizon. As is typical in real-world supervisory datasets, the distribution of $y^{(k)}$ is highly imbalanced, with failure events accounting for only a small fraction of all observations.

Finally, the objective is to learn a predictive function

$$f : \mathbb{R}^p \to [0, 1], \tag{4}$$

that estimates the probability of bank failure.

B. OVERVIEW OF THE PROPOSED FRAMEWORK

Figure 1 presents the overall architecture of the proposed framework. The primary objective is to integrate predictive performance and explainability by explicitly modeling dependencies among financial ratios and converting them into structured interaction features.

The framework comprises four main stages: data preprocessing, dependency discovery through structure learning, graph-guided feature construction, and predictive modeling with evaluation. Initially, raw financial ratios are collected for each bank and standardized to ensure comparability across variables with differing scales. Subsequently, a dependency structure is inferred from the training data using structure learning algorithms. The resulting dependency graph is then converted into an edge-level feature representation. Finally, these edge-level features are utilized independently or combined with the original node-level features to train standard predictive models.

C. DATA PREPROCESSING

Let the matrix of original financial ratios be denoted as $X \in \mathbb{R}^{n \times p}$, to ensure comparability across variables with heterogeneous scales, each feature is standardized using z-score normalization. Specifically, the normalized value of feature j for bank i is defined as

$$x_{ij}^{(\mathrm{norm})} = \frac{x_{ij} - \mu_j}{\sigma_j}, \tag{5}$$

where μ_j and σ_j denote the mean and standard deviation of feature j, respectively.

The statistics μ_j and σ_j are computed exclusively from the training set. This design prevents information leakage from validation or test data and ensures that all subsequent stages operate under realistic deployment conditions.

D. DEPENDENCY DISCOVERY THROUGH STRUCTURE LEARNING

The objective of structure learning within the proposed framework is to identify statistically significant dependency patterns among financial ratios. These patterns subsequently inform the construction of interaction features. Given the normalized feature matrix ($X \in \mathbb{R}^{n \times p}$), structure learning aims to estimate a directed graph representation of dependencies.

$$G = (V, E) \tag{6}$$

In this graph, each node V represents a financial ratio, and each directed edge E indicates a dependency inferred from the data.

In contrast to causal discovery studies, the learned graph is not interpreted as encoding causal relationships. Rather, it serves as a compact representation of recurring conditional or functional dependencies that may collectively characterize bank distress. This approach enables the application of structure learning methods without requiring strong causal assumptions, which are often difficult to justify in observational financial data.

To ensure robustness and methodological generality, four representative structure learning algorithms are employed. These span constraint-based (PC algorithm [6]), score-based (GES [16]), and continuous optimization paradigms (NOTEARS [17] and GOLEM [18]).

1) PC Algorithm

The PC algorithm is a constraint-based method that constructs a dependency graph through a sequence of conditional independence tests [6]. The process begins with a fully connected undirected graph, from which edges are iteratively removed when conditional independence is detected.

Formally, for variables X_i and X_j, the PC algorithm removes the edge between them if there exists a conditioning set $S \subseteq V \setminus \{i, j\}$ such that

$$X_i \perp\!\!\!\perp X_j \mid S. \tag{7}$$

Once the skeleton is identified, edge orientations are assigned according to established rules, resulting in a partially directed acyclic graph. Within the proposed framework, the resulting directed edges are interpreted as dependency relations, and edge weights are subsequently estimated using linear regression on the fixed structure.

2) GES

GES is a score-based structure learning algorithm that explores the space of directed acyclic graphs by optimizing a global scoring function [16]. The algorithm comprises a forward phase, which greedily adds edges to improve the score, and a backward phase, which removes unnecessary edges.

Let $\mathcal{G}$ denote the space of candidate graph structures. GES seeks to solve

$$\hat{G} = \arg\max_{G \in \mathcal{G}} \ \mathrm{Score}(G; X). \tag{8}$$

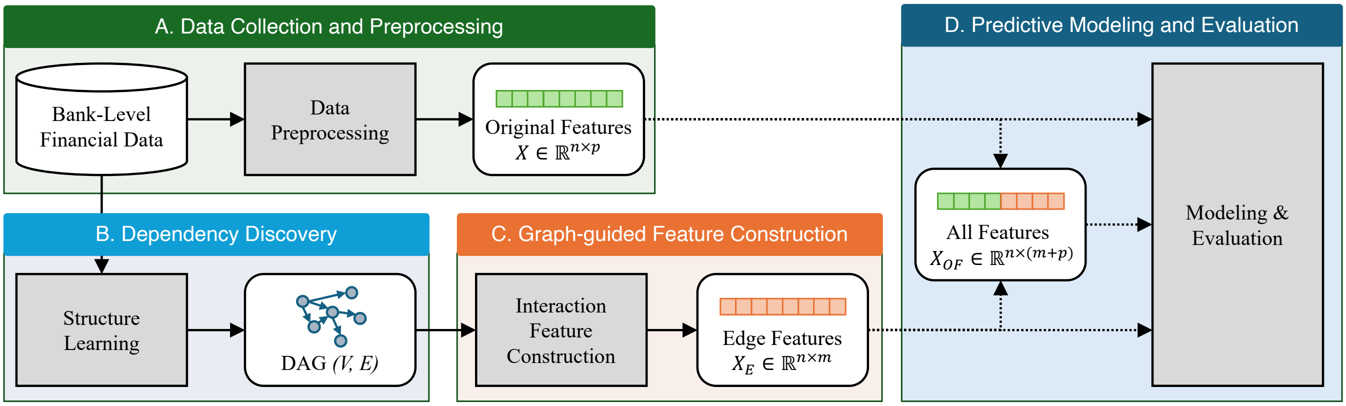


Figure 1: Overview of the proposed dependency-guided interaction feature framework

In this context, the score function is based on the Bayesian Information Criterion (BIC),

$$\mathrm{BIC}(G) = \log p(X \mid G) - \frac{\mathrm{df}(G)}{2} \log n. \tag{9}$$

The BIC score balances model fit and complexity, discouraging the formation of overly dense graphs. In this framework, GES is utilized exclusively for structure discovery, and edge weights are estimated separately to maintain comparability with other methods.

3) NOTEARS

NOTEARS formulates structure learning as a continuous optimization problem by directly parameterizing the adjacency matrix [17].

$$W \in \mathbb{R}^{p \times p} \tag{10}$$

This matrix represents weighted adjacencies, where W_{ij} indicates the strength of the directed dependency from node i to node j.

The NOTEARS objective is defined as

$$\min_{W} \ \mathcal{L}(W; X) + \lambda \|W\|_1 \tag{11}$$

subject to the acyclicity constraint

$$h(W) = \mathrm{tr}\left(e^{W \circ W}\right) - p = 0, \tag{12}$$

where $\mathcal{L}(\cdot)$ denotes a squared loss function and $\circ$ denotes the Hadamard product.

The L_1 penalty encourages sparsity in the learned graph, and the smooth acyclicity constraint ensures a directed acyclic structure. The learned weights in W are subsequently used as dependency strengths in feature construction.

4) GOLEM

GOLEM advances continuous optimization-based structure learning by explicitly modeling noise distributions and employing likelihood-based objectives [18]. It assumes a linear structural equation model

$$X = XW + Z, \tag{13}$$

where Z denotes a noise matrix, GOLEM estimates the adjacency matrix by minimizing the negative log-likelihood under DAG constraints:

$$\min_{W}; \mathcal{L}_{\mathrm{lik}}(W; X) + \lambda_1 |W|_1 + \lambda_2 h(W). \tag{14}$$

Compared to NOTEARS, GOLEM allows for node-specific noise variances and often produces more stable structures in finite-sample scenarios. As with NOTEARS, the learned adjacency matrix directly provides edge weights for feature construction.

E. GRAPH-GUIDED EDGE-LEVEL FEATURE CONSTRUCTION

Given a learned dependency graph $G = (V, E)$, the next step in the proposed framework is to transform the identified dependencies into an explicit feature representation suitable for standard predictive models.

Let $E = \{(i \to j)\}$ denote the set of directed edges identified through structure learning, and let $m = |E|$ be the number of edges. For each directed edge $i \to j \in E$, we construct an edge-level interaction feature that captures the joint activation of the source and target variables for a given observation.

Formally, for bank k, the edge-level feature associated with edge $i \to j$ is defined as follows:

$$f_{i \to j}^{(k)} = w_{ij} \cdot x_i^{(k)} \cdot x_j^{(k)}, \tag{15}$$

where $x_i^{(k)}$ and $x_j^{(k)}$ denote the normalized values of the source and target financial ratios for bank k, and w_{ij} represents the dependency strength associated with edge $i \to j$.

This formulation encodes three complementary aspects: (i) the magnitude of the source variable, (ii) the magnitude of the target variable, and (iii) the strength of the dependency inferred from the data. Consequently, the interaction feature becomes active primarily when both financial ratios assume large values in directions consistent with the learned dependency. This formulation can be interpreted as a first-order bilinear approximation of conditional activation under the learned dependency, balancing expressiveness and stability.

Aggregating across all banks and all edges produces the edge-level feature matrix

$$X_E \in \mathbb{R}^{n\times m}, \tag{16}$$

where each column corresponds to a directed dependency in the learned graph, and each row represents a bank-level observation.

In contrast to exhaustive pairwise interaction expansions, which scale quadratically with the number of variables, the proposed construction limits interactions to those supported by the learned dependency structure. This approach results in a compact and interpretable feature space, where each edge-level feature corresponds to a specific relationship between two financial ratios.

F. PREDICTIVE MODELING AND EVALUATION

The proposed framework is constructed to remain independent of the specific predictive model selected. Given a feature representation $Z \in \mathbb{R}^{n\times d}$, where d is determined by the chosen feature configuration, the framework learns a probabilistic mapping from bank-level features to failure risk.

Formally, each predictive model estimates

$$\hat{y}^{(k)} = f_\theta\left(z^{(k)}\right), \tag{17}$$

where $z^{(k)}$ denotes the feature vector for bank k, $f_\theta(\cdot)$ is a parameterized classifier, and $\hat{y}^{(k)} \in [0,1]$ represents the estimated probability of bank failure.

Model performance is evaluated along two complementary dimensions: discrimination and calibration. Discrimination metrics assess the ability to rank banks by relative risk, whereas calibration metrics evaluate the reliability of predicted probabilities. This dual perspective is essential in supervisory contexts, where both accurate risk ranking and well-calibrated probability estimates are necessary for effective decision-making.

Details regarding model selection, training procedures, and evaluation metrics are provided in the experimental setup.

IV. EXPERIMENTAL SETUP

A. DATASET AND DATA SPLITTING

The experiments are conducted on a bank-level financial dataset, where each observation corresponds to a single bank and is represented by a vector of financial ratios. Our analysis focuses on U.S. commercial banks listed on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and NASDAQ. The dataset spans from January 1969 to September 2021 and comprises 17,881 bank-year observations. All raw financial data are sourced from the COMPUSTAT database, a widely used resource in empirical banking and finance research. The target variable indicates whether a bank experiences failure within a predefined prediction horizon.

Following the early work [19], bank failure has traditionally been modeled using internal financial characteristics such as return on equity (ROE), return on assets (ROA), and operating efficiency ratios. Subsequent studies have consistently incorporated related measures of capital adequacy, asset structure, liquidity, and coverage ratios, as well as selected external conditions, to capture bank soundness and vulnerability [20]. Building on this established literature, we select a set of 12 widely used and empirically validated financial ratios. These variables span multiple categories, including solvency, profitability, efficiency, and coverage, and together provide a comprehensive snapshot of a bank's financial condition. This feature set serves as the original node-level input representation $X \in \mathbb{R}^{n\times p}$, where $p = 12$.

The dataset is partitioned into training, validation, and test sets with proportions of 70%, 10%, and 20%, respectively. The training set is used for structure learning, feature construction, and model fitting. The validation set is used exclusively for hyperparameter selection and model comparison. The test set is held out and used only for final performance evaluation.

All data preprocessing steps, including normalization, dependency discovery, and edge-level feature construction, are performed using only the training set. This design prevents information leakage from the validation or test sets and ensures that the learned dependency structures and interaction features reflect realistic deployment conditions.

B. FEATURE CONFIGURATIONS

To evaluate the contribution of graph-guided interaction features, we consider three feature configurations derived from the original financial ratios and the learned dependency graph.

- Original features (O): The original node-level financial ratios represented by the feature matrix $X \in \mathbb{R}^{n\times p}$.
- Edge-level features (F): The edge-level interaction features $X_E \in \mathbb{R}^{n\times m}$, where each feature corresponds to a directed dependency learned via structure learning.
- All features (OF): A combined representation that integrates node-level and edge-level features.

For the combined setting, the input feature matrix is constructed as

$$X_{OF} = [X \;\; X_E] \in \mathbb{R}^{n\times(p+m)}. \tag{18}$$

This augmented representation preserves the original financial indicators and enhances them with structured interaction terms informed by the learned dependency graph.

C. STRUCTURE LEARNING CONFIGURATION

Dependency structures are learned solely from the training data. We evaluate four representative structure learning algorithms: PC, GES, NOTEARS, and GOLEM. For PC and GES, the learned graph structure is fixed, and edge weights are estimated via linear regression. For NOTEARS and GOLEM, the learned weighted adjacency matrices are directly used as dependency structures.

To prevent bias in the dependency graph toward downstream predictive performance, the hyperparameters for structure learning are held constant across experiments. This ensures that differences in predictive results can be attributed to feature representations rather than overfitting in graph construction.

D. PREDICTIVE MODELS

The constructed feature representations are used to train multiple standard classifiers, including gradient boosting models, random forests, feedforward neural networks, and logistic regression. These models are selected to encompass a range of linear and nonlinear decision functions commonly applied in bank failure prediction.

For each model, hyperparameters are selected based on validation performance using a consistent search space across feature configurations. Random seeds are fixed to control for stochastic variability. No information from the test set is used during model selection or tuning.

E. EVALUATION METRICS

Model performance is evaluated on the test set using both discrimination and calibration metrics. Discriminative performance is assessed using the area under the receiver operating characteristic curve (AUROC) and the area under the precision–recall curve (AUPRC). Given the strong class imbalance, AUPRC serves as the primary metric for comparing predictive performance.

Calibration quality is evaluated using the Brier score [21] and the expected calibration error (ECE) [22]. These metrics evaluate the reliability of predicted probabilities, which is essential in supervisory and risk-monitoring applications.

V. EXPERIMANTAL RESULTS

A. SUMMARY STATISTICS

Table 1 presents summary statistics of the financial ratios used in the analysis.

The variables exhibit substantial heterogeneity in scale and dispersion. Leverage-related ratios, such as `leverage_ratio` and `debt_ebitda`, show pronounced skewness and extreme values, reflecting highly leveraged institutions. Profitability measures (`ROA` and `ROE`) include negative values, indicating periods of financial distress, while efficiency-related indicators such as `asset_turnover` and `rect_turn` display wide cross-sectional variation.

B. LEARNED DEPENDENCY STRUCTURES

In this section, we present the dependency structures learned by different structure learning algorithms and provide qualitative insights into their characteristics.

Figure 2 visualizes the dependency graphs learned using PC, NOTEARS, GOLEM, and GES, respectively. In all cases, the graphs are learned exclusively from the training data and are interpreted as dependency structures rather than causal graphs. In the graph, nodes represent financial ratios and directed edges indicate learned dependency relationships between pairs of ratios. Node colors encode structural roles: yellow nodes have both incoming and outgoing edges, green nodes have incoming edges only, red nodes have outgoing edges only, and gray nodes are isolated with no incident edges. Edge colors indicate the sign of the dependency: blue for positive relationships and red for negative relationships. Edge thickness is proportional to the absolute magnitude of the corresponding edge weight.

As presented in Figure 2(a), the dependency graph produced by the PC algorithm is sparse and conservative, containing relatively few directed edges. Several dependencies arise from leverage- and debt-related ratios, as well as profitability and capitalization measures. This structure reflects robust conditional dependency patterns, but the resulting graph exhibits limited connectivity. This sparsity indicates that the PC algorithm primarily identifies strong, direct relationships and tends to omit weaker or indirect interactions.

The graph generated by GES is the densest among the four methods, containing a greater number of directed edges and more complex interconnections, as depicted in Figure 2(b). Although this graph captures a broad spectrum of dependencies, including interactions among profitability, leverage, and liquidity ratios, the increased density may introduce redundant or weakly informative relationships. This characteristic highlights a trade-off between structural richness and parsimony inherent to score-based structure-learning approaches.

The graph generated by NOTEARS shows a more connected structure, with weighted, directed edges, as described in Figure 2(c). Debt-related ratios, such as `debt_ratio` and `totaldebt_invcap`, serve as central nodes with numerous outgoing and incoming dependencies. Compared to PC, NOTEARS produces a denser graph with graded dependency strengths, indicating its ability to capture continuous interaction patterns among financial ratios.

Lastly, the graph produced by GOLEM in Figure 2(d) exhibits a connectivity level comparable to that of NOTEARS, but with distinct differences in edge orientation and sign. Several negative dependencies are identified between capitalization-related ratios and leverage or debt indicators, which is consistent with established financial intuition. Likelihood-based formulation of GOLEM results in a structured yet flexible graph, in which both positive and negative dependencies coexist and reflect heterogeneous financial mechanisms.

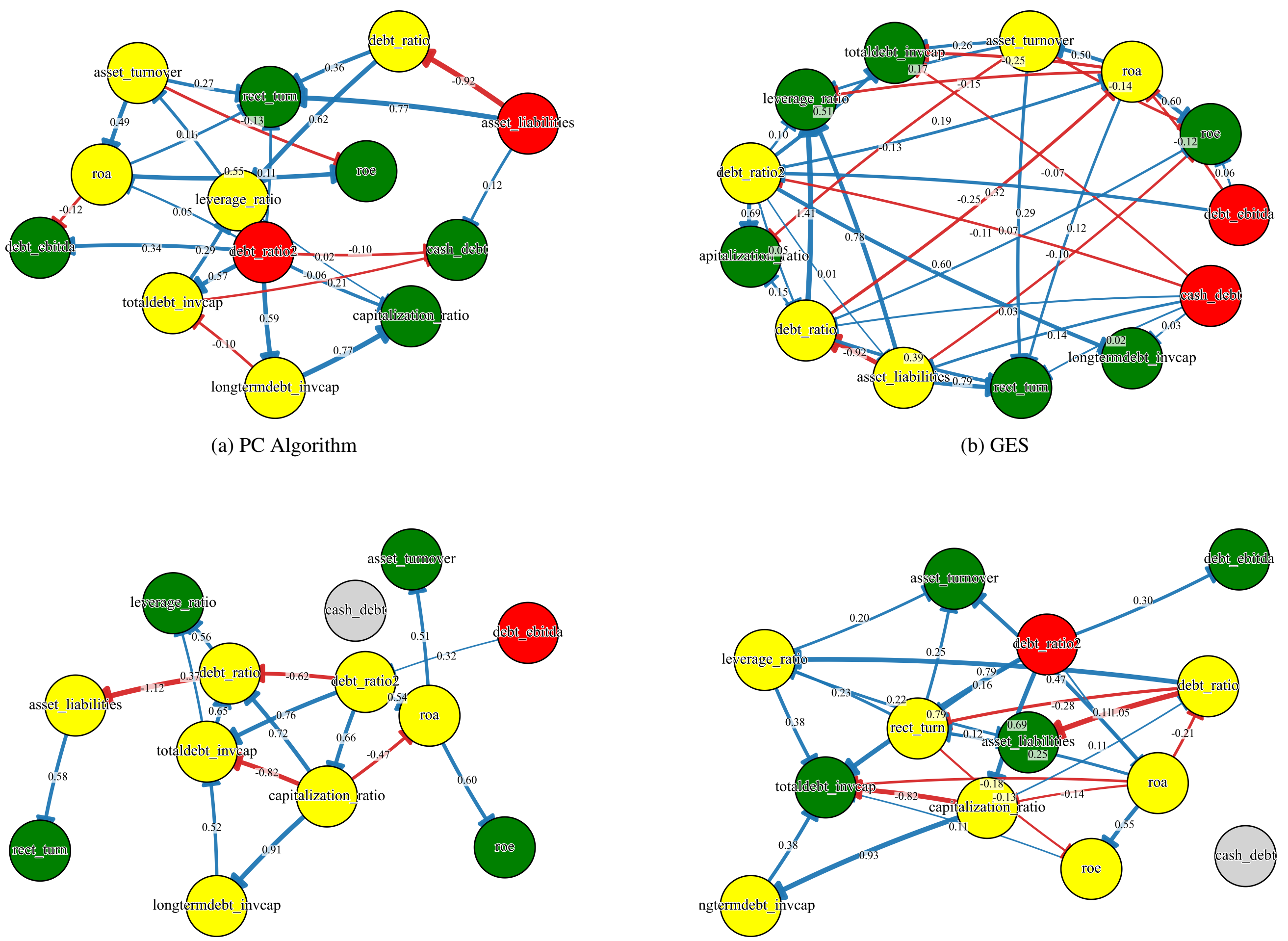
Across all methods, debt- and leverage-related ratios consistently appear as central nodes, interacting with profitability and capitalization measures. These recurring structures are consistent with established financial reasoning in bank risk analysis, where combinations of leverage, debt burden, and profitability jointly indicate financial distress.

C. PREDICTIVE PERFORMANCE

This subsection evaluates the overall predictive performance of three distinct feature configurations: (i) original node-level features (O), (ii) edge-level interaction features only (F), and (iii) the combined representation (OF). To emphasize systematic trends over model-specific effects, results are aggregated across all combinations of structure learning methods and predictive models.

Table 1: Summary Statistics of the Dataset

Variable	Mean	Std	Min	Q1	Median	Q3	Max
Leverage Ratio	10.996	4.492	0.452	8.134	10.289	13.128	101.369
Asset Liabilities	1.110	0.085	1.020	1.076	1.097	1.124	3.279
ROE	0.088	0.164	-2.379	0.062	0.104	0.145	3.531
Asset Turnover	0.075	0.036	0.019	0.054	0.072	0.087	0.918
Debt Ratio	0.904	0.044	0.311	0.890	0.911	0.929	0.978
ROA	0.023	0.015	-0.094	0.017	0.022	0.028	0.206
Capitalization Ratio	0.377	0.220	0.002	0.194	0.357	0.548	0.951
Long-term Debt / Invested Capital	0.391	0.258	0.002	0.196	0.360	0.558	3.053
Total Debt / Invested Capital	0.864	0.668	0.008	0.449	0.697	1.043	5.413
Cash / Debt	0.009	0.053	-0.535	-0.002	0.013	0.023	0.378
Debt / EBITDA	6.822	14.952	-207.980	2.463	5.109	8.987	402.721
Receivables Turnover	0.154	0.465	0.039	0.081	0.108	0.146	17.766



(a) PC Algorithm

(b) GES

(c) NOTEARS

(d) GOLEM

Figure 2: Learned dependency structures among financial ratios obtained using different structure learning algorithms.

As shown in Table 2, the original feature set (O) serves as a strong and stable baseline, reflecting the established predictive value of individual financial ratios in bank failure prediction. Across all models and structure learning methods, the O configuration achieves consistent discrimination and calibration performance.

The edge-only configuration (F) demonstrates substantially lower average performance and significantly higher variability across experimental settings. This outcome aligns with the intended design of the edge-level representation. The constructed edge features encode interactions between pairs of financial ratios but do not preserve absolute information about individual bank states. Consequently, these features are not designed to serve as standalone predictors, especially for models that rely on linear or weakly nonlinear decision boundaries. The high variance observed in the F configuration

Table 2: Overall predictive performance aggregated over all structure learning methods and predictive models.

Feature Set	AUROC	AUPRC	F1-Score	Brier	ECE
O	0.915 ± 0.026	0.381 ± 0.094	0.387 ± 0.078	0.0175 ± 0.0017	0.0103 ± 0.0038
F	0.839 ± 0.081	0.248 ± 0.157	0.289 ± 0.137	0.0193 ± 0.0023	0.0088 ± 0.0058
OF	$\mathbf{0.920 \pm 0.022}$	$\mathbf{0.389 \pm 0.099}$	$\mathbf{0.402 \pm 0.065}$	$\mathbf{0.0174 \pm 0.0017}$	0.0106 ± 0.0039

further suggests that interaction features alone are sensitive to both the underlying dependency structure and the selected predictive model.

In contrast, the combined configuration (OF) consistently outperforms the original feature set in both AUROC and AUPRC while maintaining comparable calibration performance. Notably, the standard deviation of performance metrics in the OF configuration is substantially lower than in the F configuration, indicating improved stability. These results indicate that edge-level interaction features offer complementary information that enhances predictive accuracy when combined with original financial indicators.

D. EFFECT OF STRUCTURE LEARNING METHODS

This subsection analyzes the impact of various structure-learning methods on predictive performance across multiple feature configurations. Table 3 reports the best-performing models for each structure learning method across the original (O), edge-only (F), and combined (OF) feature representations. A full comparison of predictive models is presented in Appendix A.

The original feature configuration (O) serves as a common baseline, achieving an AUPRC of 0.454 and an AUROC of 0.924.

When edge-level interaction features are used in isolation (F), performance varies substantially across structure learning methods. As shown in Table 3, the best AUPRC ranges from 0.247 for NOTEARS to 0.473 for GES, with corresponding AUROC values spanning from 0.859 to 0.928. Notably, the GES-based edge-only configuration slightly outperforms the original baseline in AUPRC (0.473), indicating that when the dependency structure is sufficiently informative, interaction features alone can capture meaningful distress signals. However, this improvement is not consistent across structure learning methods. The overall variability observed in the F configuration suggests that interaction features in isolation do not reliably surpass the original baseline. This outcome is anticipated because edge-level features capture conditional relationships among financial ratios but lack absolute information regarding individual bank states.

In contrast, the combined feature configuration (OF) consistently outperforms both O and F across all structure learning methods. Under the OF setting, AUPRC values converge within a narrow range between 0.480 and 0.489, while AUROC values remain uniformly high (0.934-0.946). Moreover, improvements are observed in F1 scores, which increase from 0.448 under O to 0.472 under GOLEM-based OF. These results demonstrate that combining edge-level interaction features with original financial ratios produces consistent and substantial improvements in predictive performance.

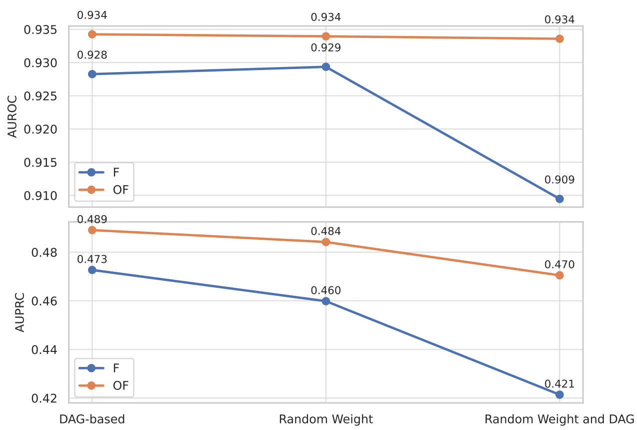


Figure 3: Robustness of predictive performance under randomization tests (GES and LightGBM)

Notably, performance differences among structure learning methods diminish under the OF configuration. While minor distinctions can be observed, such as GES achieving the highest AUPRC and NOTEARS exhibiting the most favorable calibration metrics (Brier score of 0.0156 and ECE of 0.0103), no single method consistently dominates across all evaluation criteria. This convergence suggests that the proposed framework is largely robust to the choice of dependency discovery algorithm.

E. ROBUSTNESS AND RANDOMIZATION TESTS

To ensure that the observed performance gains in the proposed framework result from meaningful dependency structures rather than incidental interactions, a series of robustness tests based on systematic randomization is conducted. The analysis focuses on the representative configuration that employs the GES-based dependency structure and the LightGBM classifier, both of which consistently demonstrated strong performance in previous experiments.

Two types of perturbations are considered. First, we randomize edge weights while preserving the learned dependency graph, thereby retaining structural connectivity but removing quantitative dependency strength. Second, we randomize both the graph structure and the edge weights, effectively destroying all learned dependency information. These perturbations are applied to both the edge-only (F) and combined (OF) configurations.

Figure 3 presents the resulting AUROC and AUPRC values under three conditions: real dependency structure, randomized edge weights, and randomized structure with randomized weights.

When only the edge weights are randomized, predictive

Table 3: DAG-level best predictive performance across feature configurations. The best-performing predictive model for each structure learning method is selected based on AUPRC.

Feature Set	DAG Method	Best Model	AUROC	AUPRC	F1	Brier	ECE
O	–	LightGBM	0.924	0.454	0.448	0.017	0.016
F	PC	LightGBM	0.908	0.424	0.432	0.018	0.017
	GES	LightGBM	0.928	**0.473**	0.452	0.017	0.016
	NOTEARS	LightGBM	0.859	0.247	0.335	0.021	0.021
	GOLEM	XGBoost	0.889	0.358	0.407	0.017	0.012
OF	PC	XGBoost	0.946	0.480	0.458	0.016	0.011
	GES	LightGBM	0.934	**0.489**	0.463	0.017	0.017
	NOTEARS	XGBoost	0.942	0.487	0.448	0.016	0.010
	GOLEM	XGBoost	0.938	0.487	0.472	0.016	0.011

performance remains largely stable. For example, the AUPRC of the F configuration decreases marginally from 0.473 to 0.460, while the OF configuration exhibits a similarly small reduction from 0.489 to 0.484. This indicates that a coherent dependency structure provides substantial predictive value, even when the precise weight magnitudes are perturbed.

In contrast, randomizing both the dependency graph and the edge weights results in a pronounced performance degradation, particularly in the edge-only configuration. In this context, the AUPRC of the F model drops sharply from 0.473 to 0.421, and the AUROC declines accordingly. This result confirms that arbitrary interaction patterns do not capture informative relationships among financial ratios. Although the OF configuration also experiences performance degradation, the magnitude of the decline is considerably smaller (AUPRC from 0.489 to 0.470), which reflects the stabilizing role of original node-level financial indicators.

F. EXPLAINABILITY ANALYSIS

To further investigate the contribution of dependency-guided interaction features to model decisions, we conduct an explainability analysis using SHAP (SHapley Additive exPlanations) [23]. The analysis focuses on the representative configuration that employs the GES-based dependency structure and the LightGBM classifier, both of which demonstrated strong and stable predictive performance in prior experiments.

Figure 4 presents SHAP summary plots for the edge-only feature configuration (F) and the combined configuration (OF), respectively.

In the F setting (Figure 4(a)), the most influential features are exclusively edge-level interaction terms, such as `asset_turnover_rect_turn` and `roa_asset_turnover`. However, the SHAP distributions exhibit substantial dispersion and sign variability, which suggests increased sensitivity to feature values and reduced stability.

In contrast, the OF configuration (Figure 4(b)) reveals a more structured and interpretable pattern of feature importance. Original node-level features such as `asset_turnover`, `asset_liabilities`, and `totaldebt_invcap` re-emerge among the top contributors, while edge-level interaction features remain highly influential. This co-existence indicates that dependency-guided interaction features serve as contextual refinements of established financial indicators rather than replacing them.

Notably, the combined model exhibits more coherent SHAP value distributions, with clearer monotonic trends between feature values and their impact on predicted failure risk. This observation aligns with the earlier robustness results, which showed that the OF configuration demonstrated greater stability under randomization.

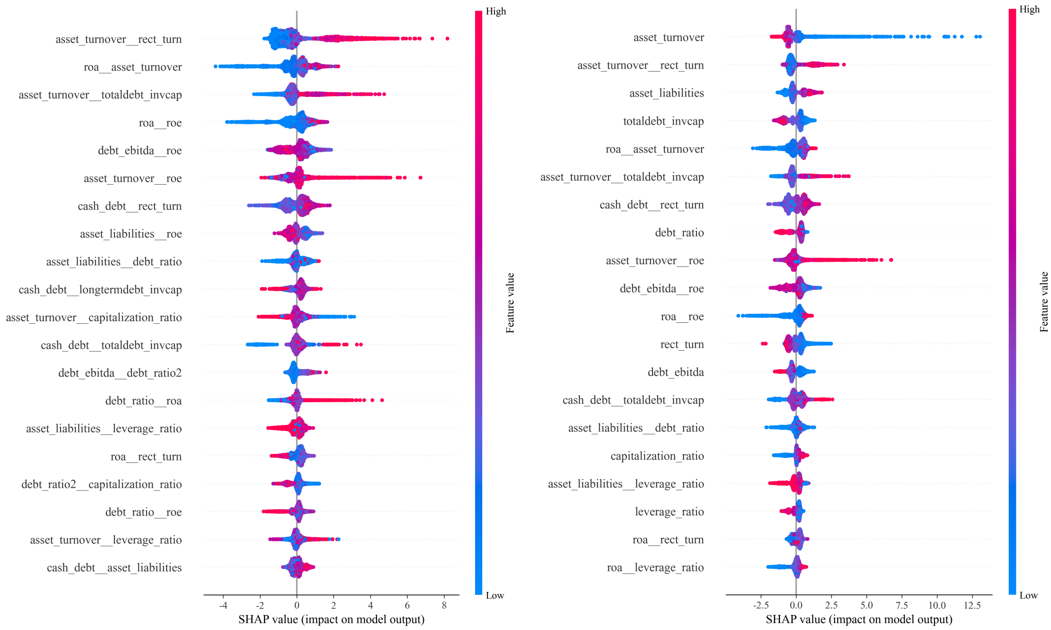
VI. DISCUSSION

A key finding of this study is that explicitly modeling dependencies among financial ratios consistently improves performance when these dependencies are used to augment, rather than substitute, the original features. Across all predictive models and structure-learning methods, the combined feature configuration (OF) demonstrates superior discrimination and stability compared to the original feature set (O), whereas the edge-only configuration (F) shows greater variance and heightened sensitivity to structural perturbations.

The comparison between O, F, and OF elucidates the distinct contributions of node-level and interaction-level information. The original configuration (O) captures the absolute financial conditions of banks, such as leverage, liquidity, and profitability levels, but treats these indicators as largely independent. In contrast, the edge-only configuration (F) encodes conditional relationships among indicators but does not retain information about absolute financial states. As a result, while F can outperform O under certain dependency structures (e.g., GES), it remains inherently unstable and susceptible to randomization, reflecting the lack of anchoring financial levels.

The OF configuration addresses this limitation by integrating both node-level and interaction-level perspectives. Original features establish a baseline representation of a bank's financial state, whereas dependency-guided interaction features serve as contextual refinements that influence the interpretation of these states. This complementary structure explains why OF consistently achieves superior and more robust performance than either O or F alone, as demonstrated by randomization tests and calibration outcomes.

The effectiveness of this approach becomes evident when reconsidering the practical emergence of financial risk signals. Most machine learning–based bank failure studies treat profitability, leverage, and liquidity ratios as independent predictors, delegating the discovery of interactions to the learn-



(a) Edge-only Features (F)

(b) Combined Features (OF)

Figure 4: SHAP summary plots for (a) edge-only features (F) and (b) combined features (OF) using the GES-based dependency structure and LightGBM classifier.

ing algorithm. However, financial distress seldom arises from isolated indicators. Risk signals typically emerge through conditional regimes. For instance, high leverage is not inherently risky when supported by strong asset turnover, but it becomes problematic under weak operational performance. Such patterns reflect interdependencies among financial ratios rather than marginal effects.

The proposed framework formalizes this intuition by employing structure learning to define a directed acyclic graph (DAG) that captures recurring dependency patterns in the data. Note that the DAG is not interpreted as a causal model. Instead, it serves as a structured hypothesis space that constrains which interactions are considered meaningful. Edge-level features are therefore not arbitrary pairwise products but are selectively activated interactions supported by the learned dependency structure.

Embedding this structured interaction mechanism into feature construction guides the predictive model toward economically plausible combinations of indicators without requiring exhaustive interaction expansion. This structural bias reduces variance, improves robustness, and enhances interpretability, as demonstrated by the stability of OF under randomization and the coherent SHAP importance patterns that jointly highlight node-level indicators and their dependency-guided interactions.

This study offers contributions from both research and practical perspectives.

From a methodological perspective, the primary contribution lies in reframing structure learning as a model-agnostic interaction generator for predictive risk modeling. The proposed framework leverages dependency structures to guide interaction modeling, thereby providing a principled method for incorporating interdependencies among financial ratios without requiring exhaustive feature expansion. In contrast to previous approaches that depend on implicit interaction learning within black-box models, this work explicitly encodes conditional relationships as edge-level features. This explicit encoding enhances both predictive performance and interpretability.

A second contribution is the systematic evaluation of multiple structure learning paradigms within a unified predictive framework. By comparing constraint-based, score-based, and continuous optimization approaches under identical experimental conditions, the study demonstrates that the benefits of dependency-guided interaction features are robust across different structure-learning algorithms. Notably, the performance gap among structure learning methods decreases substantially under the combined feature configuration. This finding suggests that the proposed framework does not depend critically on the selection of a particular dependency-discovery technique.

From a practical standpoint, the proposed approach is well aligned with supervisory and regulatory early warning systems. The framework is model-agnostic and can be integrated into existing machine-learning pipelines without modifying predictive architectures. Furthermore, the explicit representation of interactions enables risk assessments to extend beyond isolated indicator analysis toward relational explanations. This approach allows practitioners to identify problematic financial ratios and the specific combinations of conditions under which risk is amplified. This capability is particularly relevant in regulatory contexts where transparency and justification of risk signals are essential.

Despite these contributions, several limitations remain. First, the learned dependency graphs are not interpreted as causal structures, and the framework does not support counterfactual or intervention analysis. Second, the analysis is based on static financial snapshots and therefore does not capture temporal evolution or regime shifts in dependency patterns. Finally, although the proposed interaction features enhance interpretability compared to black-box models, domain expertise may still be necessary to translate complex dependency patterns into actionable supervisory insights.

Future research may address these limitations by extending the framework to dynamic or panel-based structure learning, integrating macroeconomic variables into multi-layer dependency graphs, or combining the proposed approach with causal modeling techniques to support intervention-oriented analysis. Such extensions could further enhance the applicability of dependency-guided feature construction in real-world financial monitoring systems.

VII. CONCLUSION

This study proposes a dependency-guided feature construction framework for bank failure prediction that explicitly models interdependencies among financial ratios. Extensive empirical evaluation demonstrates that dependency-guided interaction features consistently enhance predictive performance when integrated with original financial indicators. While edge-level features alone are sensitive to structural perturbations, their integration with node-level features produces robust improvements across various predictive models and structure learning methods. Randomization tests and SHAP-based explainability analysis further confirm that the observed improvements result from meaningful, non-random dependency structures and increase the interpretability of model predictions.

The findings indicate that the primary value of structure learning in bank failure prediction is its ability to organize the interaction space in a principled and data-driven manner. By constraining interaction modeling to economically plausible dependency patterns, the proposed framework reconciles predictive accuracy with robustness and transparency. This approach provides a practical means to augment existing early warning systems with structured interaction information, while maintaining model compatibility and interpretability.

In sum, this study presents a coherent framework that integrates feature-level learning with relational reasoning in financial risk modeling and identifies promising directions for incorporating structured dependency information into predictive analytics for financial supervision.

References

[1] M. Bussiere and M. Fratzscher, "Towards a new early warning system of financial crises," *journal of International Money and Finance*, vol. 25, no. 6, pp. 953–973, 2006.

[2] P. Gogas, T. Papadimitriou, and A. Agrapetidou, "Forecasting bank failures and stress testing: A machine learning approach," *International journal of forecasting*, vol. 34, no. 3, pp. 440–455, 2018.

[3] W. Zhang, "Time is the witness: Bank failure prediction with machine learning models," 2023.

[4] M. Halling and E. Hayden, "Bank failure prediction: a two-step survival time approach," *Available at SSRN 904255*, 2006.

[5] S. Gunonu, G. Altun, and M. Cavus, "Explainable bank failure prediction models: Counterfactual explanations to reduce the failure risk," *arXiv preprint arXiv:2407.11089*, 2024.

[6] T. D. Le, T. Hoang, J. Li, L. Liu, H. Liu, and S. Hu, "A fast pc algorithm for high dimensional causal discovery with multi-core pcs," *IEEE/ACM transactions on computational biology and bioinformatics*, vol. 16, no. 5, pp. 1483–1495, 2016.

[7] M. Billio, M. Getmansky, A. W. Lo, and L. Pelizzon, "Econometric measures of connectedness and systemic risk in the finance and insurance sectors," *Journal of financial economics*, vol. 104, no. 3, pp. 535–559, 2012.

[8] E. I. Altman, "Financial ratios, discriminant analysis and the prediction of corporate bankruptcy," *The journal of finance*, vol. 23, no. 4, pp. 589–609, 1968.

[9] D. Martin, "Early warning of bank failure: A logit regression approach," *Journal of banking & finance*, vol. 1, no. 3, pp. 249–276, 1977.

[10] F. Betz, S. Oprică, T. A. Peltonen, and P. Sarlin, "Predicting distress in european banks," *Journal of Banking & Finance*, vol. 45, pp. 225–241, 2014.

[11] J. Kolari, D. Glennon, H. Shin, and M. Caputo, "Predicting large us commercial bank failures," *Journal of Economics and Business*, vol. 54, no. 4, pp. 361–387, 2002.

[12] D. G. Mayes and H. Stremmel, *The effectiveness of capital adequacy measures in predicting bank distress*. SUERF studies, 2014, no. 2014/1.

[13] T. Beck, A. Demirgüç-Kunt, and M. S. M. Pería, "Bank financing for smes: Evidence across countries and bank ownership types," *Journal of Financial Services Research*, vol. 39, no. 1, pp. 35–54, 2011.

[14] N. Bussmann, P. Giudici, D. Marinelli, and J. Papenbrock, "Explainable machine learning in credit risk management," *Computational Economics*, vol. 57, no. 1, pp. 203–216, 2021.

[15] A. Famà, J. Myftiu, P. Pagnottoni, and A. Spelta, "Explainable machine learning for financial risk management: two practical use cases," *Statistics*, vol. 58, no. 5, pp. 1267–1282, 2024.

[16] D. M. Chickering, "Optimal structure identification with greedy search," *Journal of machine learning research*, vol. 3, no. Nov, pp. 507–554, 2002.

[17] X. Zheng, B. Aragam, P. K. Ravikumar, and E. P. Xing, "Dags with no tears: Continuous optimization for structure learning," *Advances in neural information processing systems*, vol. 31, 2018.

[18] I. Ng, A. Ghassami, and K. Zhang, "On the role of sparsity and dag constraints for learning linear dags," *Advances in Neural Information Processing Systems*, vol. 33, pp. 17 943–17 954, 2020.

[19] P. A. Meyer and H. W. Pifer, "Prediction of bank failures," *The journal of finance*, vol. 25, no. 4, pp. 853–868, 1970.

[20] N. Alzayed, R. Eskandari, and H. Yazdifar, "Bank failure prediction: corporate governance and financial indicators," *Review of Quantitative Finance and Accounting*, vol. 61, no. 2, pp. 601–631, 2023.

[21] K. Rufibach, "Use of brier score to assess binary predictions," *Journal of clinical epidemiology*, vol. 63, no. 8, pp. 938–939, 2010.

[22] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran, "Measuring calibration in deep learning." in *CVPR workshops*, vol. 2, no. 7, 2019.

[23] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in neural information processing systems*, vol. 30, 2017.

CHIWOO LEE is currently pursuing his B.Sc in the School of Information Convergence at Kwangwoon University. His research interests encompass machine learning, financial modeling, and the application of artificial intelligence.

SUJIN PYO received his Ph.D. degree in the Department of Industrial Engineering at Seoul National University in 2018. His research interests include finance, industrial artificial intelligence, large language model and financial connectedness. He has published papers in the field of finance, employing diverse statistical models and AI techniques.

MINSU CHO (Member, IEEE) received his Ph.D. degree in the School of Management Engineering at Ulsan National Institute of Science and Technology (UNIST) in 2018. He is currently an associate professor with the School of Information Convergence at Kwangwoon University. His research interests include process mining, natural language processing, business analytics, and industrial artificial intelligence. He has published more than 30 scientific papers in several top-tier venues, including Decision Support Systems, Expert Systems with Applications, Journal of Biomedical Informatics, and International Journal of Medical Informatics.

APPENDIX A COMPLETE EXPERIMENTAL RESULTS

Table 4: Complete predictive performance results across all feature configurations, structure learning methods, and predictive models.

Feature Set	DAG Method	Model	AUROC	AUPRC	F1	Brier	ECE
O	–	Logit	0.924	0.255	0.259	0.019	0.011
		FFMLP	0.871	0.303	0.369	0.020	0.008
		RF	0.920	0.440	0.443	0.016	0.005
		XGBoost	0.938	0.450	0.415	0.016	0.011
		LightGBM	0.924	0.454	0.448	0.017	0.016
F	PC	Logit	0.690	0.043	0.076	0.022	0.004
		FFMLP	0.761	0.075	0.141	0.021	0.004
		RF	0.920	0.389	0.425	0.017	0.004
		XGBoost	0.914	0.410	0.390	0.017	0.012
		LightGBM	0.908	0.424	0.432	0.017	0.017
F	GES	Logit	0.749	0.085	0.190	0.022	0.004
		FFMLP	0.806	0.141	0.165	0.021	0.007
		RF	0.928	0.423	0.413	0.016	0.003
		XGBoost	0.936	0.448	0.422	0.016	0.012
		LightGBM	0.928	0.473	0.452	0.017	0.016
F	NOTEARS	Logit	0.709	0.049	0.081	0.021	0.003
		FFMLP	0.762	0.072	0.137	0.021	0.007
		RF	0.852	0.216	0.289	0.019	0.007
		XGBoost	0.861	0.252	0.324	0.019	0.011
		LightGBM	0.859	0.247	0.335	0.021	0.021
F	GOLEM	Logit	0.741	0.064	0.148	0.022	0.003
		FFMLP	0.779	0.097	0.129	0.021	0.004
		RF	0.903	0.334	0.408	0.017	0.005
		XGBoost	0.889	0.358	0.407	0.017	0.012
		LightGBM	0.890	0.358	0.405	0.019	0.019
OF	PC	Logit	0.922	0.261	0.383	0.019	0.012
		FFMLP	0.883	0.275	0.347	0.020	0.009
		RF	0.928	0.436	0.444	0.016	0.005
		XGBoost	0.946	0.480	0.458	0.016	0.011
		LightGBM	0.932	0.479	0.484	0.016	0.016
OF	GES	Logit	0.924	0.259	0.273	0.019	0.012
		FFMLP	0.884	0.294	0.400	0.020	0.008
		RF	0.922	0.440	0.400	0.016	0.007
		XGBoost	0.941	0.481	0.472	0.016	0.012
		LightGBM	0.934	0.489	0.463	0.017	0.017
OF	NOTEARS	Logit	0.925	0.259	0.327	0.019	0.012
		FFMLP	0.880	0.289	0.293	0.020	0.009
		RF	0.903	0.434	0.381	0.016	0.004
		XGBoost	0.942	0.487	0.448	0.016	0.010
		LightGBM	0.935	0.478	0.450	0.016	0.016
OF	GOLEM	Logit	0.922	0.265	0.322	0.019	0.013
		FFMLP	0.880	0.290	0.352	0.020	0.008
		RF	0.930	0.436	0.402	0.016	0.004
		XGBoost	0.938	0.487	0.472	0.016	0.011
		LightGBM	0.935	0.468	0.464	0.017	0.016